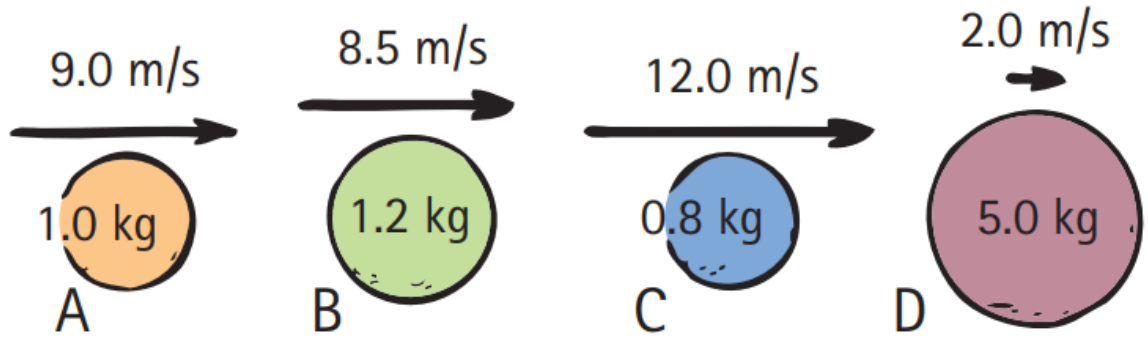
Trắc nghiệm

Q1.

Ý tưởng gần giống câu này:

69. The balls have different masses and speeds. Rank the following from greatest to least: (a) Momentum. (b) The impulses needed to stop the balls.



Q2.

How much work is done on a **250-kg crate** that is hoisted **2 m** in a time of **4 s**?

A. 7000 J
B. 5000 J
C. 2000 J
D. 500 J

Q3.

The acceleration of a vertically-thrown ball at the top of its path is

A. between zero and 10 m/s^2
B. 10 m/s^2
C. zero
D. dependent on the mass of the ball

Q4.

When no air resistance acts on a projectile, its horizontal acceleration is

A. centripetal
B. at right angles to g
C. g
D. zero

Q5.

A freight train rolls along a track with considerable momentum.
If it then rolls at the same speed but has **twice as much mass**, its momentum is now

A. unchanged
B. four times as large
C. twice as large
D. zero

Q6.

A girl pushes a cart to the left with a **180-N** force.
At the same time a boy pushes it to the right with a **75-N** force.
The net force exerted on the cart is:

→ 105 N to the left

Q7.

If both the mass and velocity of an object remain constant, the impulse on the object is

A. zero
B. equal to its momentum
C. equal to its kinetic energy
D. increasing with time

Q8.

Which of the following is a unit of force?

A. kg
B. $kg \cdot m/s^2$
C. kg/s
D. m/s^2

Q9.

You drop a pillow off the edge of the administration building on your campus. While the pillow is falling, its speed

A. and acceleration both increase
B. and acceleration both decrease
C. increases and its acceleration decreases
D. decreases and its acceleration increases

Q10.

A 2-kg ball rolling at 3 m/s bumps into a pillow and comes to a stop in **0.5 s**.
What average force does the ball exert on the pillow?

A. 6 N
B. 12 N
C. 3 N
D. 24 N

Q11.

Can we use the equation $F_{\text{net}} = \frac{dp}{dt} = ma$ when the mass is constant?

A. No, because the given equation is applicable for the variable mass only.
B. No, because the given equation is not applicable for the constant mass.
C. Yes, and the resultant equation is $F = ma$.
D. Yes, and the resultant equation is $F = mv$.

Q12.

The speed of a vertically-thrown ball at the top of its path is

A. dependent on the mass of the ball.
B. 10 m/s.
C. between zero and 10 m/s.
D. zero.

Q13.

For how long should a force of 50 N be applied to change the momentum of an object by 48 kg·m/s?

A. 2 s
B. 0.96 s
C. 16.6 s
D. 2400 s

Q14.

A 120-N falling object encounters 40 N of air resistance. The net force on it is

A. 80 N
B. 160 N
C. 40 N
D. 120 N

Q15.

When a 15-kg block is simultaneously pushed eastward with 50 N and westward with a force of 15 N, the combination of these forces on the block is

A. 35 N west
B. 35 N east
C. 65 N east
D. 65 N west

Q16.

When you toss a ball upward with a certain initial speed, it reaches a maximum height h after it leaves your hand.
If you throw the ball upward with double the initial speed, the maximum height the ball reaches is

A. $h/2$
B. $8h$
C. $4h$
D. $2h$

Q17.

Which pair of quantities contains both a scalar and a vector?

A. acceleration and momentum
B. charge and resistance
C. kinetic energy and mass
D. temperature and velocity

Q18.

A freely falling object has a speed of 15 m/s at one instant. Exactly 1 s later its speed will be

A. 40 m/s
B. 30 m/s

C. 20 m/s
D. 25 m/s

Q19.

The reason why a 20-kg rock falls no faster than a 1-kg rock in free fall is that the

A. force of gravity is the same for both.
B. 20-kg rock has greater acceleration.
C. force/mass ratio is the same for both.
D. 1-kg rock has greater acceleration.

Q20.

Considering Newton's third law, why don't two equal and opposite forces cancel out each other?

A. Because the two forces act on different systems.
B. Because the two forces have different magnitudes.
C. Because the two forces are applied at different points in space.
D. Because one force is always weaker than the other.

Tự luận

An airplane is flying horizontally with a speed of **1000 km/h (280 m/s)** when one of its engines falls off.
Neglect air resistance, and assume that it takes **30 s** for the engine to hit the ground.

(a) Show that the altitude of the airplane is **4.4 km**.
(Use $g = 9.8\ m/s^2$.)

(b) Show that the horizontal distance the engine travels during its fall is **8.4 km**.

(c) If the airplane continues to fly forward at the same speed as though nothing had happened, where is the engine relative to the airplane at the moment it hits the ground?